

Sensor Q&A

Q1: Which parameter does `./prog_vpe` take to realize sensor graphing?

You need to bring an ini file later. This ini file should be selected corresponding to the connected sensor interface, and the configuration in the ini file should be modified according to your usage scenario. For specific usage, please refer to the vpedemo usage documentation in the sdk.

Q2: Sensor is abnormal @sensor pad 0?

You can first check the dts configuration of the sensor to see if the vif mode and csi lane configuration are correct, and refer to the sensor configuration document developed by sigdoc BSP.

Q3: Under the dts configuration of sensor, SR0 itself does not reuse BT1120, why is there `vif_sr0_bt1120_mode = <1>`?

dts is the description passed to the lower-level software to initialize the hardware. If there is one, the corresponding mode will be configured. If it is not, it will have no effect.

Q4: Multiple sensor pads are connected to the same type of sensor. How to load the sensor driver to be compatible with these pads at the same time?

Set by the chmap parameter, chmap is designed to make the board compatible with different sensors, and it is calculated by bit mask. For example, both sensor pad 0 and pad 1 are connected to imx307, then use `ismod imx307_MIPI.ko chmap=3 lane_num=4`.

Q5: An error occurs in the SNR pad enable fail program?

First, confirm whether the interface is correct and the wiring is in good contact according to the hardware guide of the demo board. Then check whether the chmap and lane_num parameters

loaded by the driver are set correctly. Finally, if the configuration file is read, check the settings of parameters such as sensorpad in the ini file.

 Q6: Which demo reference is used to obtain raw data and capture functions?

To get it from vif, you can use the interface of get vif buffer, MI_SYS_ChnlInputPortGetBuf of the mi sys interface.

 Q7: The compression mode in the MI_SYS_FrameData_s attribute of the vif output port is E_MI_SYS_COMPRESS_MODE_TO_8BIT. How to get the uncompressed raw data from the vif?

Turning off fbc is uncompressed data

```
echo 0 > /sys/module/mi_vif/parameters/bFBCMode
echo FBC 0 > /proc/vpe/vpe_debug
```

 Q8: When using the vpe program to debug the bt1120 of the nvp6158c, is it possible to directly load the nvp6158c.ko file (insmod /config/modules/4.9.227/nvp6158c.ko chmap=2 lane_num=4), and then run the prog_vpe program? What needs to be modified in the param.ini file?

Use the sr1 group of pins to connect to the bt1120 input, you need to use sensor pad 1, group 3, vifdev 12 to receive data, and you need to modify VifDev = 12 in the param ini file.

 Q9: If the data sent by the sensor to the main chip is in yuv format, can I skip the isp and directly bind other modules after vif?

You can skip the isp module when you don't need to use isp, and just bind other modules directly.

 Q10: BT1120 outputs an output not finished timeout error, how to troubleshoot?

cat /sys/class/mstar/vif0/vif_info and vif_ints, if the vif information is all 0, it means that the vif has not received data, check whether the dts is paired and whether the BT1120 input waveform is correct.

 Q11: If VIF uses 2MUX, should the param.ini file set VifDev = 12 and VifDev = 13, RunMode = 2Multi? For 4MUX, the param.ini file sets VifDev = 12 and VifDev = 13, VifDev

= 14, VifDev = 15, RunMode = 4Multi?

There is no need to set this up. There are 4 devices in a group, just use the group configuration. For example, group 0 sets VifDev = 12, RunMode = 2Multi, and uses VifDev12 and VifDev13 by default.